Free response of a spring-mass-damper system

MIE 402 | ME Laboratory II | Fall 2026 | Prof. Ehsan Roohi

Measure the spring stiffnesses, then record how a cart oscillates and settles for seven configurations. Use Vernier Graphical Analysis for data collection and MATLAB for the spring and cart calculations.

1. Weigh and label the equipment

1. Measure and record the mass of the **cart**, each **hex rod**, and the parts of the **magnetic damper** that move with the cart. Record all values in kg (100 g = 0.100 kg).

2. For every cart run, calculate the total moving mass: cart + installed hex rods + all damper components that move. Fixed parts on the track are not included.

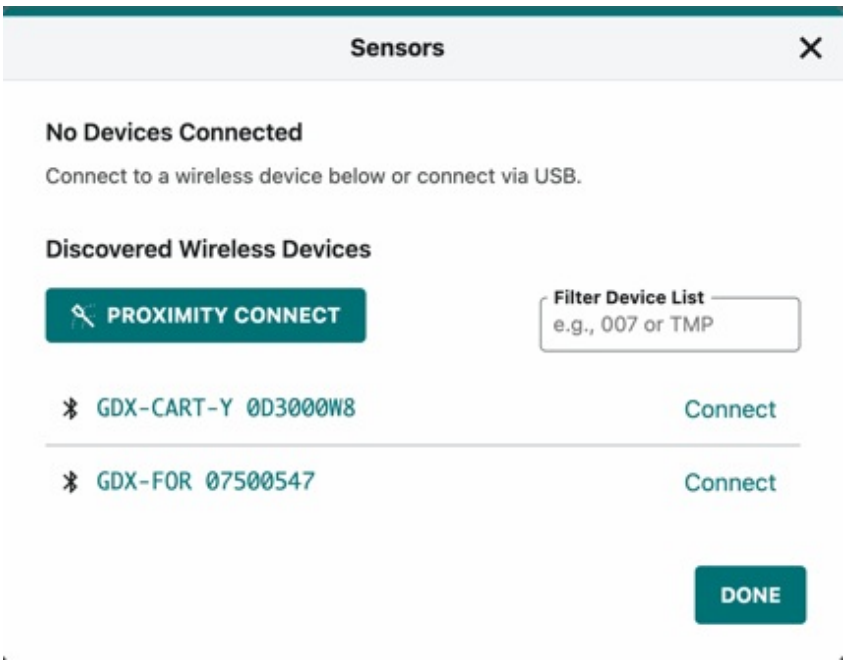
3. Label the four springs A-D. For the hanging-spring test, measure the total hanging mass, including the hanger and attached moving pieces. This is a separate mass from the cart mass.

Record for every run: spring IDs, cart mass plus moving attachments, number of magnets, initial displacement, sampling rate and CSV filename.

2. Connect the Vernier sensor

1. Turn on the sensor. Open **Vernier Graphical Analysis → New Experiment → Sensor Data Collection**, or click the sensor connection icon.

2. Match the serial number on your device to the list, click **Connect**, then **Done**. Select the **force sensor** for the hanging-spring test; connect the **cart** for the cart test.



Vernier sensor list: connect the device assigned to your station by its serial number.

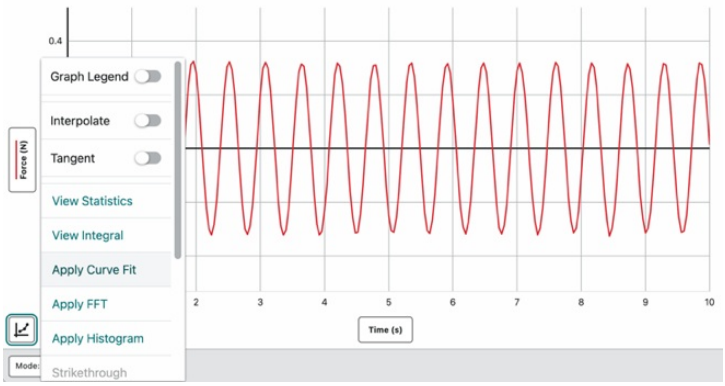
Keep the track clear and do not overstretch the springs. Ask the GTA if a sensor or attachment is loose.

Measure the stiffness of the four springs

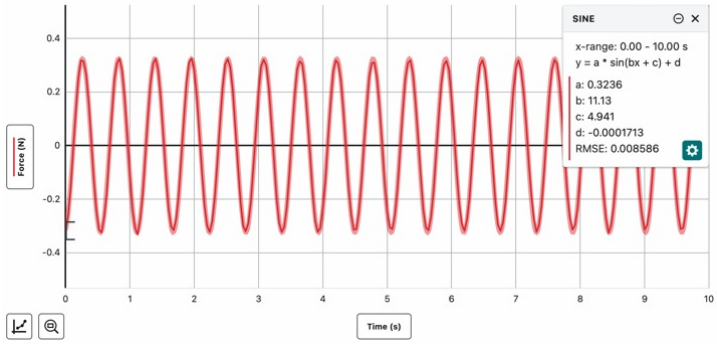
1. Secure the force sensor vertically on the stand. With its hook unloaded, choose **Force → Zero**. Hang spring A and a known mass from the hook. Record the total hanging mass, including the hanger.

2. Display **Force (N) versus Time (s)**. In the collection settings, use 25 samples/s and 5-10 s as a starting point. Click **Collect**, pull the hanging mass down slightly, and release gently. Keep its motion vertical.

3. Select several clean cycles after release. Open the graph tools at the lower left → **Apply Curve Fit → Sine**. Check that the curve follows the measured data. Record the fitted **B** value and save a screenshot of the fit.



Open graph tools and choose Apply Curve Fit.



Choose Sine and read B in the fit panel. This image is an example.

In the Vernier fit **F = A sin(Bt + C) + D**, B is angular frequency in rad/s. C is the phase. Under the light-damping approximation, the spring stiffness is:

$$k \approx m_h B^2$$

Here k is spring stiffness [N/m] and m_h is the total hanging mass [kg]. For example, m_h = 0.200 kg and B = 7.30 rad/s give k = 10.66 N/m. Use your own measurements.

4. Export the force record as CSV: document/file menu → **Export → CSV → Save CSV**. Save one file per spring. Repeat the measurement and sine fit for springs B, C and D.

Calculate k in MATLAB; prepare the cart

3. Spring stiffness analysis in MATLAB

1. On the Lab 4 website, download and unzip **Spring analysis package**. Keep both .m files in the same folder. Open **Run_Lab4_Spring_Stiffness.m** in MATLAB and click **Run**.

2. Choose where to save results and enter **4** for the number of spring records. For each spring, enter its ID and total hanging mass. Choose **1 = Enter Vernier B**, enter B and select **rad/s**. If the software equation reports f in Hz instead, select Hz. Enter 0 for alpha when using the light-damping approximation.

3. Alternatively, choose **2 = Import Force-Time CSV/Excel**. Select the time and force columns and a clean interval after release. Inspect the measured force, fitted curve and residual. Save the results for each spring.

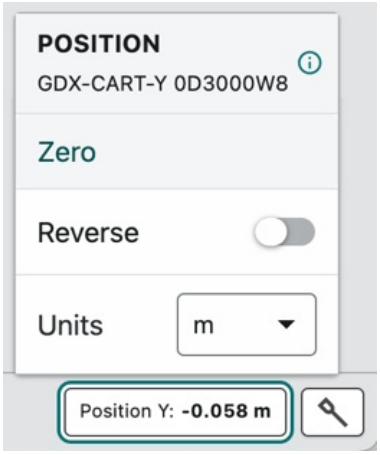
4. Open **Spring_Stiffness_Summary.csv** in the results folder and record k_A, k_B, k_C and k_D [N/m]. Save the Vernier fit screenshots. When entering B manually, the MATLAB program calculates k but has no measured force data to plot.

4. Assemble and zero the cart

1. Place the cart on a level track between two labeled springs. Attach the springs securely and keep both taut during the motion; their lengths should be about 15 cm near equilibrium, as in the course setup. Record which spring is on each side.

2. Turn on the cart and connect it in Vernier by serial number. Display **Position (m) versus Time (s)**. Let the cart come to rest at equilibrium.

3. Click the **Position** readout and choose **Zero** while the cart is at equilibrium. Move the cart a known short distance and verify that the position changes; return it to equilibrium. If the cart is lifted or repositioned, zero it again at equilibrium.



Position menu: choose Zero with the cart at rest at equilibrium.

> **Do not zero at the release position.** After zeroing at equilibrium, move the cart to the desired starting displacement and then release it.

Collect the cart tests and save your data

5. Record seven configurations

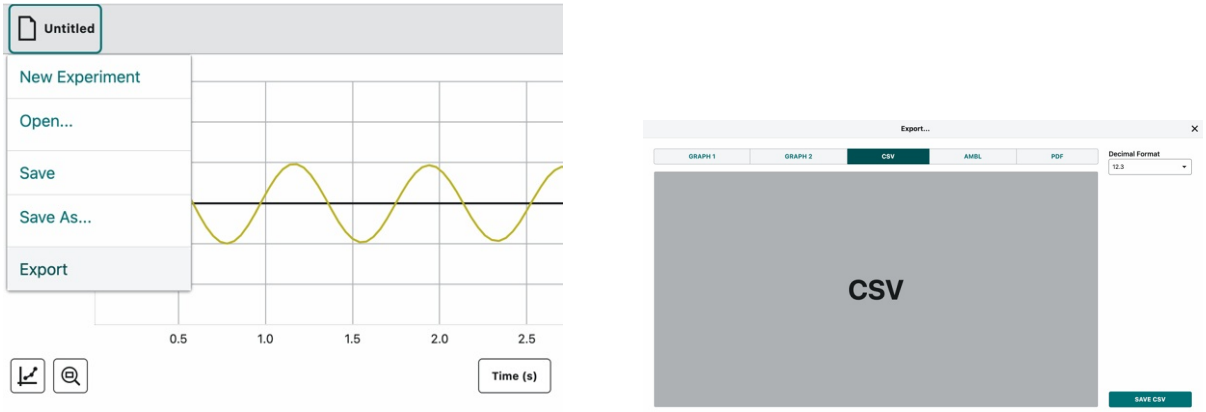
Case	Configuration	Record
1	Base cart, spring pair, no magnets	Total moving mass and spring IDs
2-3	Add one hex rod, then a second	Total cart + rod mass for each run
4-5	Base springs; use 2, then 4 magnets	Number of magnets and total moving mass, including damper parts
6-7	Base mass; two other spring pairs; no magnets	Spring IDs and measured stiffness of each spring

1. For each case, check that both springs remain taut and the cart can move without hitting the end stops. Record the total moving mass and configuration. Change one factor at a time.

2. With the cart at rest at equilibrium, check or set **Position → Zero**. Set the sampling rate to 25 samples/s or less according to the laboratory setup, and choose a duration long enough to capture several cycles and the decay.

3. Move the cart a measured small distance from equilibrium. Click **Collect** first, then release without pushing. Stop after the motion has decayed or before the cart reaches an end stop.

4. Open the document/file menu → **Export → CSV → Save CSV**. Use a different filename for every case, such as **Group2_Run04_Magnets2.csv**. Open the CSV in Excel or MATLAB to confirm that time and position data were saved. Then clear the displayed run and repeat.



Vernier export: choose Export (left), select CSV and save the file (right).

Before leaving: save four spring stiffness values, all seven cart CSV files and the metadata table. Analyze one cart case and show its plot and results to the GTA.

Analyze the cart and prepare the report

6. Analyze each cart run in MATLAB

1. Download the separate cart analysis script **MIE402_Lab4_Analyze_Free_Response.m** from the Lab 4 site and click **Run**. Select a cart position CSV.

2. Enter the total moving mass [kg], the measured left spring stiffness k_L [N/m], the measured right spring stiffness k_R [N/m], and the number of magnets. Enter the two individual spring values; the program calculates $k_{eq} = k_L + k_R$.

3. Inspect the raw position plot. Enter the start and end of the free motion after release, and select the position unit used in your CSV. Check the selected peaks and plots, then save the generated figures and summary.

7. What to include in the combined Labs 4-5 report

Spring measurements: a table of the four spring IDs, hanging masses, fitted B values and calculated k [N/m], with representative Vernier fit evidence.

Cart tests: a table for all seven cases with spring IDs, total moving mass, magnet count, measured frequencies and damping results. Include clear position-versus-time and frequency-spectrum plots with axes, units and case labels.

Comparison: compare the measured frequencies with the prediction using your measured mass and springs. Explain how changing mass, spring stiffness and magnets changes the motion. Include only equations used in the analysis.

Keep the original CSV files and MATLAB output. Check Canvas for report submission details.

Quick troubleshooting

Problem	Check
No spring oscillation in Vernier	Connect the force sensor, select Force (N), and check the hanging mass and sensor connection.
Unexpected k value	Check hanging mass units, include the hanger, and confirm B is in rad/s. Fit several clean cycles.
Cart position does not change	Connect the cart and select Position; check that the wheel rolls on the track. Zero again at equilibrium.
MATLAB cannot analyze the file	Select the correct spring Force CSV or cart Position CSV, check units and select an interval after release. Ask the GTA if needed.

Lab 4 website: instructions, spring analysis package and cart analysis script.